PSEUDOLINE ARRANGEMENTS, CONTINUED: CELLS, SKELETA AND ADDED PSEUDOLINES, AFTER GROTHENDIECK'S WORKING NOTES

MICHEL HUA

Abstract. In notes from 1981 to 1984, Grothendieck studied arrangements of pseudolines in the real projective plane through their cell decompositions and their one-dimensional skeleta. A companion paper proves the bigon-free dichotomy of these notes. This one treats three further statements.

(1) Page 26 of folder 80 states that the closed cells of an arrangement meet pairwise in a single closed cell or not at all exactly when no pseudoline is bigonal and the arrangement is not four lines in general position. This is false, already for five straight lines. The correct criterion, which is the hypothesis of the theorem on page 24, is that no two points carry every pseudoline between them. We prove it for line arrangements, and in covector form for every symmetric set of sign vectors closed under composition.

(2) Page 54 asserts that a bigon-free arrangement is determined, functorially, by its skeleton together with the decomposition of its edges into pseudolines. For combinatorial maps we prove faithfulness, a lifting criterion, the reading of triangle orientations on the skeleton, and fullness whenever every triangle has a fourth pseudoline avoiding its vertices, in particular whenever every triangle has a vertex on only two pseudolines. The remaining configurations, among them the complete quadrangle, are reduced to a sign lemma on chirotopes and the topological representation theorem.

(3) Folder 154 asserts that the position of an added pseudoline is determined by the cyclic order in which it meets the others, and that the stable positions near a given one are the vertices of a cube. Both statements hold for single-element extensions of rank-3 oriented matroids.

The results of (1) and (3), the statements on maps in (2) and the sign lemma on chirotopes are formalised in Lean; the topological representation theorem is assumed throughout.

1. Introduction

An *arrangement of pseudolines* is a finite family of at least three simple closed non-separating curves in the real projective plane $\mathbb{RP}^2$, any two of which meet once, where they cross, not all through one point [4, 2]. It decomposes $\mathbb{RP}^2$ into vertices, edges and faces. The University of Montpellier holds the mathematical papers that Grothendieck left to Jean Malgoire and publishes their facsimiles. Its folder 80, *Systèmes de pseudo-droites*, undated and placed "from 1981" by the inventory, and its folder 154, catalogued as *[Système de pseudo-droites]*, whose dated pages run from December 1983 to February 1984, study these arrangements as cell complexes. A companion paper [5] proves the boxed dichotomy of folder 80 on bigon-free arrangements, those in which every pseudoline carries at least three vertices. Here we treat three other statements of the two folders. A statement is attributed to Grothendieck when the page states it, and is otherwise said to be *after* the page; the proofs are ours. Section 5 lists what is formalised in Lean. The topological representation theorem of Folkman and Lawrence [3] is used but not formalised.

We use the names of the companion paper: II_4 and II_5 are four and five pseudolines in general position; $\mathrm{II}'_{3,2}$ is three pseudolines through a point p and two through a point q,

Date: 7 November 2026.

2020 *Mathematics Subject Classification.* 52C30; 52C40, 05C10, 57M15.

Key words and phrases. pseudoline arrangements, oriented matroids, covectors, chirotopes, generalised maps, cell decompositions, single-element extensions, Lean.

ORCID 0009-0002-8304-0683.

without the pseudoline joining p and q; III_6 is the complete quadrangle. The *two-pencil* types I_N, $\mathrm{I}_{p,q}$, $\mathrm{II}'_{p,q}$ of page 22 are the arrangements with two points p, q such that every pseudoline passes through p or through q.

2. When closed cells meet in a cell

Pages 24 and 26 call an arrangement *latticial* when the closed cells of its decomposition of $\mathbb{RP}^2$ meet pairwise in the empty set or in a single closed cell. Then the closed cells, ordered by inclusion and with a least and a greatest element adjoined, form a lattice. Page 26 states, without proof:

> Proposition (Grothendieck, folder 80, p. 26). An arrangement is latticial iff (a) no pseudoline is bigonal and (b) it is not of type II_4.

The word "bigonal" is a doubtful reading of the manuscript, and the page writes "pas de type II_4". We read (b) as "is not II_4", not "contains no II_4": the second reading would exclude II_5, which page 24 calls latticial. The "only if" half is correct: a bigonal pseudoline, or II_4, gives two closed cells whose intersection is a pair of vertices. The "if" half is false.

Proposition 2.1 (the proposition of page 26 fails)**.** *The arrangement $\mathrm{II}'_{3,2}$ of the five lines $x = 0$, $x = z$, $x = 2z$ (through $[0 : 1 : 0]$) and $y = 0$, $y = z$ (through $[1 : 0 : 0]$) has no bigonal line and is not II_4, but it is not latticial. Its two corner faces meet exactly in the two centres $[0 : 1 : 0]$ and $[1 : 0 : 0]$, which do not form one cell.*

Each of the five lines carries at least three vertices. In the affine chart $z = 1$ (Figure 1), the corners $x > 2, y > 1$ and $x < 0, y < 0$ are glued through the line at infinity into one face F, and the corners $x < 0, y > 1$ and $x > 2, y < 0$ into one face G. The closures of F and G use different edges of each line; for instance, the parts $y > 1$ and $y < 0$ of the line $x = 0$ are two edges separated by the vertex $[0 : 1 : 0]$. So the closures meet in the two centres only. Page 24 lists $\mathrm{II}'_{3,2}$ (written $\mathrm{I}'_{3,2}$ there) among the types it calls latticial, so that remark fails too. The correct criterion is the hypothesis of the theorem of page 24.

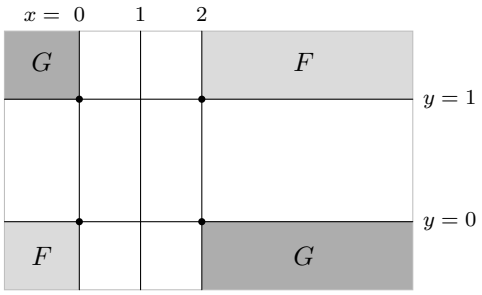


Figure 1. Proposition 2.1 in the affine chart $z = 1$. The lines $x = 0, 1, 2$ meet at $[0 : 1 : 0]$, at the top and bottom of the chart; the lines $y = 0, 1$ meet at $[1 : 0 : 0]$, at its left and right. Through the line at infinity, which is not in the arrangement, the two light corners form one face F of $\mathbb{RP}^2$ and the two dark corners one face G. Their closures share no edge and meet exactly in the two centres at infinity.

Theorem 2.2 (after folder 80, pp. 24 and 26)**.** *An arrangement of lines in $\mathbb{RP}^2$ is latticial iff it is not of two-pencil type, that is, iff there are no two points such that every line passes through one of them.*

The proof works in the covector model. A line arrangement in $\mathbb{RP}^2$ comes from vectors $\ell_1, \dots, \ell_n$ in $\mathbb{R}^3$. A point $x \in \mathbb{R}^3 \setminus \{0\}$ has the sign vector $X = (\operatorname{sign}\langle \ell_i, x\rangle)_i$, and the cells of the sphere are the sets of points with a given sign vector. The closure of the cell of X is the union of the cells of the sign vectors *conformal* to X, that is, equal to X wherever they are non-zero. The cells of $\mathbb{RP}^2$ are the pairs $\pm X$, and the closed cell of $\pm X$ is identified with

the set of non-zero covectors conformal to X or to $-X$. The set $\mathcal{L}$ of covectors is symmetric ($-X \in \mathcal{L}$ for $X \in \mathcal{L}$) and closed under *composition*, $(X \circ Y)_i = X_i$ if $X_i \neq 0$ and Y_i otherwise.

Theorem 2.3 (covector form)**.** *Let $\mathcal{L}$ be a finite symmetric set of sign vectors closed under composition. The closed cells of $\mathcal{L}/\pm$ meet pairwise in the empty set or in a single closed cell iff there are no two non-zero $v, w \in \mathcal{L}$ with disjoint supports.*

Proof. If v and w have disjoint supports, the closed cells of $v \circ w$ and $v \circ (-w)$ both contain $\pm v$ and $\pm w$. Every covector in their intersection is, up to sign, a face of v or of w. A single closed cell equal to the intersection would be the closed cell of such a covector, and its support could not contain the supports of both v and w. Conversely, let X, Y be non-zero covectors. If some non-zero covector is a face of both X and $-Y$, replace Y by $-Y$. Then no non-zero face u of X is a face of $-Y$: otherwise u and the face just found have disjoint supports. So the common faces of the closed cells are, up to sign, the common faces of X and Y. These are closed under composition. Let M be one with the largest support. For every common face Z, the composite $M \circ Z$ is again a common face and has support at least that of M, hence the same support, so Z is a face of M. The intersection is the closed cell of M, or empty if there is no common face. □

For lines, two non-zero covectors with disjoint supports are two points such that every line passes through one of them, which gives Theorem 2.2. For pseudolines, the topological representation theorem [3] identifies the cell decomposition with the covectors of a rank-3 oriented matroid [1], and Theorem 2.3 gives the same criterion; that step is not formalised.

3. The skeleton determines the arrangement

Pages 39 to 54 of folder 80 study the functor that sends a bigon-free arrangement to its *decomposed skeleton*: the graph of vertices and edges, with the partition of the edges into pseudolines. Page 39 states full faithfulness for skeleta equipped with an orientation of every triangle. Pages 41 to 53 try to recover the arrangement and these orientations from the skeleton, under conditions (A1) to (A6), and page 54 concludes:

> Theorem (Grothendieck, folder 80, p. 54). The canonical functor from bigon-free arrangements of pseudolines to bigon-free one-dimensional arrangements of pseudolines is fully faithful.

In particular a bigon-free arrangement is determined by its decomposed skeleton, and every automorphism of the decomposed skeleton comes from the arrangement. The pages leave the proof incomplete. Page 46 says that (A1) to (A3) hold "bien entendu" for a skeleton that comes from an arrangement; none of (A1) to (A6) is proved. We work, as page 60 suggests, with *combinatorial* arrangements. Such an arrangement is a generalised map [7] (flags with involutions $\alpha_0, \alpha_1, \alpha_2$) of Euler characteristic 1 whose flags are labelled by pseudolines. Each pseudoline is connected, passes once through each of its vertices and *goes straight* there: half a turn of the rotation $\alpha_2\alpha_1$ around a vertex returns to the same pseudoline. Every vertex lies on at least two pseudolines, two pseudolines meet at exactly one vertex, and every pseudoline carries at least three vertices.

Figure 2 shows the data used below. A *corner* at a vertex is a pair of consecutive edges around it. A *three-edge face path* is a corner continued along its second edge to the next corner of the same face.

Theorem 3.1 (after folder 80, pp. 41–54)**.** *For bigon-free combinatorial arrangements:*

1. (Faithfulness.) *Two isomorphisms of arrangements inducing the same maps on vertices and on edges are equal.*
2. (Page 41, lifting.) *An isomorphism of decomposed skeleta comes from an isomorphism of arrangements iff it preserves, in both directions, the three-edge face paths.*
3. (Pages 42 and 50–51, orientation of a triangle.) *In homology modulo* 2, $\dim Z_1/B_1 \leq 1$*; this is the relation $c_2 = c(K)$ of page 42. The cycle of a pseudoline bounds nothing,*

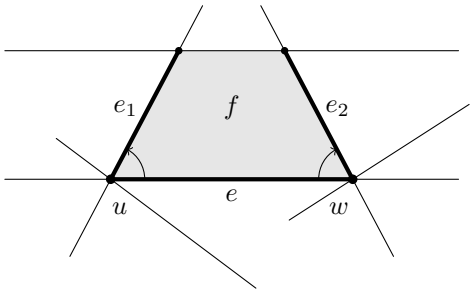


FIGURE 2. A three-edge face path (e_1, e, e_2) around the face f: the corner (e_1, e) at the vertex u, through which three pseudolines pass, continued along e to the corner (e, e_2) at w. Around u the six edge-ends alternate between the three pseudolines, and half a turn returns to the same pseudoline.

and a pseudoline that crosses the boundary of a union of faces transversally crosses it an even number of times. Consequently, let three non-concurrent pseudolines form a triangle and let Δ be a fourth pseudoline through none of their three meeting points. A triangle cycle, made of one segment of each side between the meeting points, bounds iff Δ crosses an even number of its segments. This is condition (A5) of page 51; it is read on the skeleton and does not depend on Δ.

(4) (Page 54, fullness.) *If every triangle has a fourth pseudoline avoiding its three vertices, every isomorphism of decomposed skeleta lifts, uniquely, to an isomorphism of arrangements. This is the case when every triangle has a vertex on exactly two pseudolines, in particular for simple arrangements.*

Sketch. (1) A flag d and the flag $\alpha_2 d$ have the same vertex and the same edge. Their images under α_1 lie on different edges, because the arrangement is bigon-free. So a map of flags commuting with α_1 is determined by what it does on vertices and edges. (2) The three-edge face paths are the rotation system of the map, the data (19) to (21) of page 41. (3) Euler characteristic 1 and connectedness give the dimension bound. A pseudoline goes straight, so in half a turn around each of its vertices it meets every other pseudoline through that vertex once. This gives the parity statement, and shows that, for a cycle z and the cycle D of a pseudoline, exactly one of z and $z + D$ bounds. Of the eight triangle cycles, the four that Δ crosses an even number of times are those that bound. (4) The sides of an edge are read through bounding triangles, by (3), when a fourth pseudoline avoids the triangle; this gives (A1) to (A4) at vertices of any multiplicity. Two edges at a vertex are adjacent iff no edge lies between them. So the corners, hence the face paths, are read on the skeleton, and (2) applies. A triangle with a vertex on only two pseudolines always has a fourth pseudoline avoiding its vertices. □

What remains are the triangles whose three vertices are multiple and through which every other pseudoline passes, the case of condition (A6) on pages 52 and 53; the complete quadrangle III_6, on page 53, is one of them (Figure 4). Page 51 opens this "2ème cas" by supposing first that three pseudolines D_a, D_b, D_c through the vertices a, b, c can be chosen not concurrent, and draws that configuration; when they are concurrent at a point d, the six pseudolines ab, bc, ca, ad, bd, cd form the complete quadrangle III_6. We do not follow the page's configuration, and handle both subcases at once with oriented matroids. A *chirotope* assigns a sign to each triple of pseudolines without common point, the orientation of the triangle they form [1]. The following lemma uses only incidences and alternation.

Theorem 3.2 (the edition's sign lemma)**.** *Consider finite sets of points and lines with an incidence relation such that two distinct lines meet in exactly one point, every point lies on at least two lines, and every line carries at least three points. Call* basis *a triple of distinct lines without common point. Let χ and χ' be alternating sign functions on the bases. Suppose that*

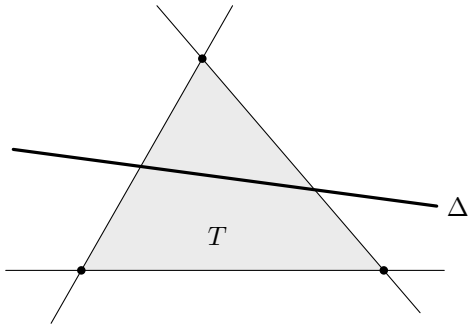


FIGURE 3. Condition (A5). The pseudoline Δ avoids the three vertices of the triangle and crosses two of the three segments bounding the face T, an even number, and this cycle bounds T. Replacing one segment of ∂T by the complementary segment of the same pseudoline changes the parity, since Δ meets that pseudoline once: the new triangle cycle bounds nothing.

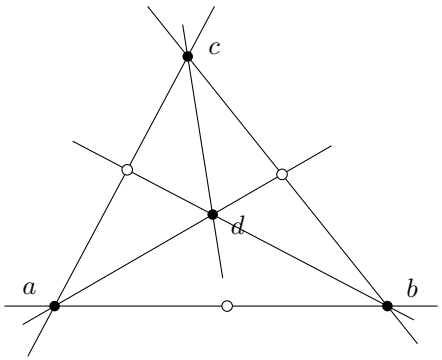


FIGURE 4. The complete quadrangle III_6, which page 24 calls the tetrahedral arrangement: the six lines through four points a, b, c, d in general position, with $4 + 3 = 7$ vertices, the four triple points (black) and the three diagonal points (white). The triangle of the lines ab, bc, ca has three triple vertices, and each of the other lines passes through one of them.

for every line i and all lines j, k, l, m such that (i, j, l), (i, j, m), (i, k, l), (i, k, m) are bases,

$$\chi(i,j,l)\,\chi(i,j,m)\,\chi(i,k,l)\,\chi(i,k,m) = \chi'(i,j,l)\,\chi'(i,j,m)\,\chi'(i,k,l)\,\chi'(i,k,m).$$

Then there are signs r_i such that $\chi'(i,j,l) = r_i r_j r_l\,\chi(i,j,l)$ for every basis: χ' is a reorientation of χ.

Sketch. Fix a line i. The other lines fall into classes, one per point of i, and the hypothesis says that the function $(j,l) \mapsto \chi(i,j,l)\,\chi'(i,j,l)$ has product 1 around every cycle of four edges of the complete multipartite graph on these classes. A sign function with this property on a complete multipartite graph with at least three parts is a constant times a cut; a line has at least three points, so there are at least three parts. Hence $\chi'(i,j,l)\,\chi(i,j,l) = s_i\,\varepsilon_i(j)\,\varepsilon_i(l)$ for some signs s_i and $\varepsilon_i(j)$. The function $\psi(i,j,l) = \chi'(i,j,l)\,\chi(i,j,l)\,s_i s_j s_l$ is symmetric and has, at each line i, the form $\psi(i,j,l) = \delta_i(j)\,\delta_i(l)$ with $\delta_i(j) = \varepsilon_i(j)\,s_j$. It remains to show $\psi = 1$ on every basis; then $r = s$. If a fourth line avoids the three points of a basis (i,j,l), multiplying the relations at i, j and l for the four lines gives $\psi(i,j,l) = 1$, as in (A5). Otherwise cut the triangle a, b, c along a line x through $b \cap c$ and a third point of a. The triangles a, b, x and a, x, c each have a fourth line, through $a \cap c$ and $a \cap b$ respectively. In the complete quadrangle the auxiliary lines x are the sides through the fourth triple point. □

On an actual arrangement, the product $\chi(i,j,l)\chi(i,j,m)\chi(i,k,l)\chi(i,k,m)$ is -1 exactly when the vertices $i \cap j$ and $i \cap k$ separate $i \cap l$ and $i \cap m$ on the pseudoline i. This is a

statement about the rank-2 contraction at i; we do not formalise it. An isomorphism of decomposed skeleta preserves the incidences and, along each pseudoline, the cyclic order of the vertices up to reversal. Transporting the chirotope of the second arrangement to the first, Theorem 3.2 shows that the two chirotopes differ by a reorientation. The topological representation theorem [3] then gives a homeomorphism of $\mathbb{RP}^2$ carrying one arrangement to the other. It induces the given isomorphism of skeleta: it respects the pseudolines and the vertices, and since a bigon-free pseudoline carries at least three vertices, an edge is determined by its pseudoline and its two ends. For topological arrangements, these steps give the theorem of page 54 for every bigon-free arrangement, modulo the topological representation theorem and the rank-2 statement above. The argument is ours and has not been checked by a specialist. For arrangements of straight lines, Goodman and Pollack proved a statement of the same kind: two arrangements of lines are isomorphic iff their circular sequences of permutations are locally equivalent, that is, each line switches with the others in the same or in the opposite order [2, Th. 5.2.2].

4. An added pseudoline

Folder 154 studies the positions of a pseudoline added to an arrangement Σ. In oriented-matroid terms a position is a *single-element extension* of the rank-3 oriented matroid of Σ. Such an extension is given by its signature σ on the cocircuits, which are the vertices of Σ, with $\sigma(-Y) = -\sigma(Y)$: the sign records on which side of the new pseudoline the vertex lies, or 0 if the vertex is on it. By a theorem of Las Vergnas [6], [1, Th. 7.1.8], a signature defines a single-element extension iff it does so on every rank-2 minor. In rank 3 we use it line by line. Choose a representative of each vertex and read the signs along each pseudoline of Σ, corrected by the orientation of the pseudoline. Then σ is a *position* when, along each pseudoline, these signs read $+\cdots+\ (0)\ -\cdots-$ up to reversal, with at most one 0, and are not all 0. The last condition excludes the pseudolines of Σ themselves.

Page 18 asserts that the position is determined by the cyclic order, compatible with antipody, in which the double cover of the added pseudoline meets the oriented pseudolines of Σ ("il ne devrait pas être difficile de montrer"). In oriented-matroid terms that order is the rank-2 contraction at the new element.

Theorem 4.1 (after folder 154, p. 18)**.** *Two single-element extensions of a rank-3 chirotope whose contractions at the new element agree up to a global sign are equal, up to that sign on the triples containing the new element.*

Proof. The triples not containing the new element are those of Σ. By alternation, a triple containing it is, up to the sign of a permutation, a value of the contraction, or 0 if it contains the new element twice. □

The proof is elementary in this form. What page 18 needs beyond it is the dictionary between pseudolines and chirotopes, which is again the topological representation theorem. Page 150 introduces a cellular surface X^* of positions. Page 153 describes the positions near a given one D through ν vertices of Σ: the stable positions that generalise D correspond to the vertices of a cube of dimension ν, the sub-stable ones to its edges, incidence being specialisation, with 2^ν vertices and $\nu\,2^{\nu-1}$ edges. Say that τ is *more generic* than σ, or a *generisation* of σ, when $\tau = \sigma$ wherever $\sigma \neq 0$.

Theorem 4.2 (after folder 154, p. 153)**.** *Let σ_D be a position vanishing on exactly ν vertices of Σ. A pseudoline of Σ carries at most one of these vertices, and every sign vector more generic than a position is a position. Consequently, restriction to the ν zeros of σ_D is a bijection from the generisations of σ_D onto all sign vectors on these zeros, and specialisation corresponds to specialisation. These sign vectors are the faces of the cube $[-1,1]^\nu$, ordered by reverse inclusion. There are 3^ν generisations of D, of which 2^ν are stable (no zero, the vertices of the cube) and $\nu\,2^{\nu-1}$ sub-stable (exactly one zero, its edges).*

Proof. A valid sequence $+\cdots+\ (0)\ -\cdots-$ has at most one zero, so two zeros of σ_D never lie on a common pseudoline of Σ. A generisation changes only zeros, and replacing the zero of a valid sequence by either sign keeps it valid, so a generisation of a position is a position. Since the zeros lie on different pseudolines, the choices at the ν zeros are free and independent. The counts follow. □

The proof uses neither the hypothesis $\nu \geq 2$ of the page nor the axioms of an arrangement. The relation with the surface X^* of page 150 is not treated here. Figure 5 shows $\nu = 2$ and $\nu = 3$. The page's own sketches support this reading of ν as the number of vertices of D: a chain of two spindles on a pseudoline carries one arrow at each vertex, a choice of side, and the bottom of the page draws the cube for $\nu = 3$. The top sketch labels pseudolines through a single vertex Δ_1, Δ_3, Δ_ν, an indexing we could not reconcile with the text's Δ_s, indexed by vertices; that label may be a misreading.

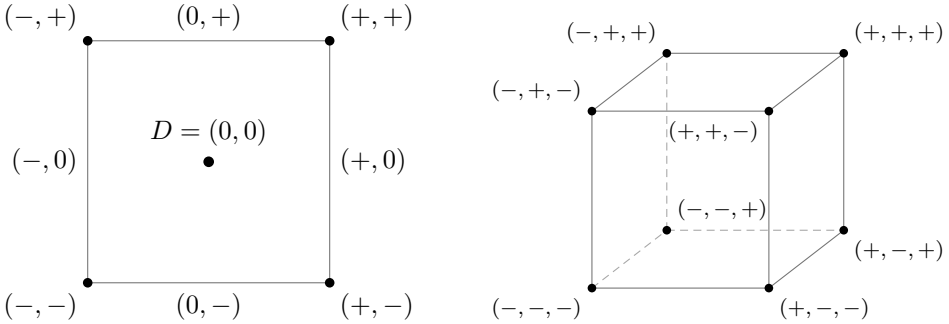


FIGURE 5. Theorem 4.2 for $\nu = 2$ (left) and $\nu = 3$ (right): the generisations of a position D through ν vertices, labelled by their signs on these vertices. On the left D is the square itself, drawn at its centre; the 4 stable positions are its corners and the 4 sub-stable ones its edges. On the right the $2^3 = 8$ stable positions are the corners of the cube and the $3 \cdot 2^2 = 12$ sub-stable ones its edges; the faces and D, the cube itself, are not labelled. The bottom of page 153 draws this case as a projected cube and a circle with eight marks.

5. VERIFICATION

The following are formalised in Lean 4 with Mathlib, in the repository Commutator-IO/grothendieck-project, with only Lean's standard axioms. Section 2: `Folder80Maps1.lean` (`lattice_iff`, `lattice_iff_not_twoPencil`, `closure_cell`, `II32_not_lattice`, `lattice_criterion_false`, `not_lattice_of_bigonal`, `not_lattice_of_four`). Theorem 3.1: `Folder80Maps3.lean` (`faithful`, `lift_iff`, `finrank_Z1_le`, `lineChain_not_mem`, `parity`, `triangle_iff`, `full_simple`, `full_of_fourth'`, `fourth_at`, `full_of_simpleCorner`). Section 4: `Folder154.lean` (`problem17_holds`, `genEquiv_spec_iff`, `cubeFace_subset_iff`, `card_gen`, `card_gen_stable`, `card_gen_substable`).

Theorem 3.2: `Folder80Signs.lean` (`const_add_cut`, `eq_zero_of_fourth`, `eq_zero`, `locallyEquivalent_of_sameSeparation`, `reorientation_of_locallyEquivalent`, `reorientation_of_sameSeparation`), with the second sign function transported along a bijection of the lines and alternation assumed on bases only. An earlier version of that file assumed alternation on all triples, degenerate ones included, with values in $\mathbb{Z}/2$; no function satisfies that, so the assembled theorems held vacuously. The hypotheses are now restricted to bases, as in the proof above, and the theorems are proved as stated. Also not formalised: the topological representation theorem of Folkman and Lawrence, hence the passage from topological pseudoline arrangements to covectors, maps and chirotopes; the rank-2 reading of separation on a pseudoline; the surface X^* of folder 154, page 150.

Statement on the use of artificial intelligence

The transcriptions of folders 80 and 154, the modernised readings, the proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the statements, the proofs and the references, and takes full responsibility for the content. The formal proofs are checked mechanically by Lean.

Acknowledgements

The facsimiles are published by the University of Montpellier, grothendieck.umontpellier.fr.

References

[1] A. Björner, M. Las Vergnas, B. Sturmfels, N. White, G. M. Ziegler, *Oriented Matroids*, 2nd ed., Encyclopedia of Mathematics and its Applications 46, Cambridge University Press, 1999, doi:10.1017/CBO9780511586507.

[2] S. Felsner, J. E. Goodman, Pseudoline arrangements, in J. E. Goodman, J. O'Rourke, C. D. Tóth (eds.), *Handbook of Discrete and Computational Geometry*, 3rd ed., CRC Press, Boca Raton, FL, 2017, ch. 5.

[3] J. Folkman, J. Lawrence, Oriented matroids, *J. Combin. Theory Ser. B* 25 (1978), 199–236, doi:10.1016/0095-8956(78)90039-4.

[4] B. Grünbaum, *Arrangements and Spreads*, CBMS Regional Conference Series in Mathematics 10, American Mathematical Society, 1972.

[5] M. Hua, A bigon-free dichotomy for pseudoline arrangements, after a note of Grothendieck, preprint, 2026, `https://grothendieck.commutator.io/article/grothendieck-bigon.pdf`.

[6] M. Las Vergnas, Extensions ponctuelles d'une géométrie combinatoire orientée, in *Problèmes combinatoires et théorie des graphes* (Orsay, 1976), Colloq. Internat. CNRS 260, CNRS, Paris, 1978, 265–270.

[7] P. Lienhardt, N-dimensional generalized combinatorial maps and cellular quasi-manifolds, *Internat. J. Comput. Geom. Appl.* 4 (1994), 275–324, doi:10.1142/S0218195994000173.

Commutator, Paris, France
Email address: `michel@commutator.io`